

BLOCKCHAIN INVESTIGATION REPORT

1. Case Details

- Case ID: CASE-2024-ETH-0891
- Investigation ID: INV-9042-SIH
- Investigation Date: 2024-03-15 14:30:00 UTC
- Investigator: Cyber Crime & Intelligence Forensics Unit
- Blockchain: Ethereum (Mainnet)
- Suspect Wallet: 0x47ac0fb4f2d84898e4d9e7b4dab3c24507a6d503
- Report Status: COMPLETED (MULTI-HOP TRACED)

2. Investigation Summary

Metrics Overview

- Total Transactions Analyzed: 11 transfer events across 7 distinct wallets
- Total Incoming Value: 30,000,000.0 DAI + 125.40 ETH
- Total Outgoing Value: 30,000,000.0 DAI + 0.055 ETH
- Number of Hops Traced: 3 (Full Breadth-First Traversal)
- Destination Addresses Identified: 10 unique Ethereum destination nodes
- Overall Risk Level: HIGH
- Confidence Score: 94%

Investigator Executive Summary

Automated blockchain multi-hop tracing initiated on victim-reported suspect Ethereum wallet **0x47ac0fb4f2d84898e4d9e7b4dab3c24507a6d503** identified immediate structured fund dispersion. In Hop 1, 30,000,000 DAI stablecoins were moved to primary relay wallet **0xf977...acec** alongside contract triggering calls. In Hop 2, funds were split and routed into token liquidity and relay nodes (including 70,000,000 MFT tokens to **0x3f5c...f0be** and secondary stablecoin interactions). In Hop 3, secondary nodes dispersed assets into native ETH micro-transactions and USDC liquidity pools (**24,948.96 USDC** to **0x5777...2168**). Graph traversal completed with 0 infinite loops detected via automated cycle prevention.

3. Transaction Evidence

Hop	Transaction Hash	Date/Time (UTC)	Amount & Asset	Category / Indicator
1	0x2e46c5a5fd5b...	2021-06-03 02:04:05	30,000,000.0 DAI	ERC-20 (Large Volume)
1	0x2e46c5a5fd5b...	2021-06-03 02:04:05	0.0 ETH	External (Contract Call)
2	0x9fc4f7d1b040...	2019-07-09 14:54:02	70,000,000.0 MFT	ERC-20 (Token Relay)
2	0x31f60d794bbc...	2022-08-31 16:32:51	100.0 daiblack.com	ERC-20 (Airdrop/Spam)
3	0xd7be6ba52109...	Block 4116404	0.045 ETH	External (Native ETH)
3	0x902ff3a220cf...	2022-08-20 22:58:31	24,948.96 USDC	ERC-20 (Stablecoin Pool)

Transaction Flow Summary



4. Risk Summary

• Risk Score: 82 / 100 | Risk Level: **HIGH** | Confidence Score: 94%

Risk Indicator	Status	Key Evidence
High transaction frequency	TRUE	Burst outgoing transactions executed in identical and adjacent blocks.
Large-value movement	TRUE	Single transactions exceeding 30,000,000 DAI stablecoins.
Rapid movement through wallets	TRUE	Funds propagated through 3 successive intermediary hops.
Repeated splitting	TRUE	Primary balance divided into multiple secondary destination paths.
Repeated consolidation	FALSE	No downstream funneling to a single consolidation hub detected within 3 hops.
Suspicious address interaction	TRUE	Interaction with unverified token contracts and phishing airdrop signatures.
Mixer/Bridge interaction	FALSE	No known privacy mixer (e.g. Tornado Cash) directly in the analyzed 3-hop tree.

5. VASP Attribution

VASP Interaction Detected: The investigation identified direct linkage between the subject wallet and recognized Virtual Asset Service Provider (VASP) hot wallet infrastructure.

- **VASP Name:** Binance Exchange
- **Associated Address:** 0x47ac0fb4f2d84898e4d9e7b4dab3c24507a6d503
- **Interaction Type:** Originating Hot Wallet / Outgoing Fund Dispersion
- **Transaction:** 0x2e46c5a5fd5b52f66a0b7d97998a70cfefd5ce944e999fe5ac296a1a0a171f20
- **Attribution Confidence:** 98%

Interaction Analysis: The investigation identified a direct link between the suspect starting address and recognized exchange deposit/withdrawal infrastructure. Legal preservation notices may be served to obtain KYC records.

6. Key Findings and Investigator Assessment

Critical Findings

- High-volume DAI stablecoin dispersion (30M DAI) routed immediately upon receipt.
- Three-tier multi-hop relay graph without immediate user-level holding periods.
- Secondary conversion into USDC stablecoin liquidity pools (**24,948.96 USDC** at Hop 3).
- Automated cycle detection successfully prevented infinite loops during graph traversal ($A \rightarrow B \rightarrow A$).

Final Assessment

The transaction patterns strongly indicate automated fund dispersion and structuring across intermediate Ethereum wallets. The initial liquidity originates from an identified exchange hot wallet and rapidly distributes across multiple secondary addresses. High priority for subpoena and preservation requests.

7. Recommended Actions

- **Extend Transaction Tracing:** Deepen BFS tracing beyond 3 hops for the USDC liquidity branch (address `0x5777d92f208679db4b9778590fa3cab3ac9e2168`).
- **Preserve Evidence:** Preserve all on-chain transaction hashes and block timestamps for formal legal proceedings.
- **Serve VASP Subpoena:** Issue formal preservation request to Binance Exchange for account metadata associated with transaction `0x2e46c5a5...` .
- **Monitor High-Risk Relays:** Place wallet alerts on intermediate addresses `0xf977...acec` and `0xed76...764b` for subsequent outgoing flows.

8. Limitations

- **Pseudonymity:** Blockchain addresses are pseudonymous and do not directly prove real-world legal identity without verified off-chain VASP attribution or KYC subpoenas.
- **Transaction Scope:** Analysis is limited to on-chain data within the selected 3-hop traversal depth and the 100-transfer-per-node rate limit.
- **Off-Chain Activities:** Off-chain internal ledger transactions occurring within centralized exchanges are not reflected on the public Ethereum ledger.
- **Fund Continuity Nuance:** In account-based blockchains like Ethereum, multi-hop paths ($A \rightarrow B \rightarrow C$) prove transaction relationships and directional fund flows, but do not mathematically guarantee that the exact same atomic tokens received from A were forwarded to C, due to potential fund commingling or pre-existing balances in intermediary wallets.
- **Mixers & Cross-Chain Bridges:** The use of mixers, decentralized privacy pools, or cross-chain bridges can obscure downstream traceability.
- **Analytical Indicators:** Risk scores and confidence ratings are forensic heuristics and do not constitute a definitive legal finding of guilt.

9. Variable Dictionary

Variable Identifier	Definition & Forensic Meaning
{{CASE_ID}}	Unique law enforcement or internal investigative case number.
{{SUSPECT_WALLET}}	Target 42-character Ethereum hex address under investigation.
{{BLOCKCHAIN}}	Target blockchain ecosystem (Ethereum Mainnet).
{{TRANSACTION_COUNT}}	Total number of transfer relationships analyzed across all hops.
{{TOTAL_INCOMING}}	Aggregated gross value received by the subject wallet.
{{TOTAL_OUTGOING}}	Aggregated gross value transferred out of the subject wallet.
{{NUMBER_OF_HOPS}}	Maximum traversal depth reached during Breadth-First Search.
{{RISK_SCORE}}	Algorithmic risk score between 0 and 100 based on transaction indicators.
{{OVERALL_RISK_LEVEL}}	Categorical classification: LOW, MEDIUM, or HIGH.
{{CONFIDENCE_SCORE}}	Statistical confidence percentage in findings and data attribution.

10. Conditional Logic Table

Condition Trigger	Rule Status	Report Action Taken
{{HIGH_TRANSACTION_FREQUENCY}}	TRUE	Display high-frequency transactional burst finding in Section 4.
{{RAPID_MOVEMENT}}	TRUE	Include rapid multi-hop propagation finding in Section 4 & Section 6.
{{REPEATED_SPLITTING}}	TRUE	Display fund dispersion and splitting graph analysis in Section 3 & 4.
{{SUSPICIOUS_INTERACTION}}	TRUE	Flag unverified token contract interactions in Section 4.
{{MIXER_OR_BRIDGE_INTERACTION}}	FALSE	Omit mixer deanonymization section; mark mixer interaction as FALSE.
{{VASP_INTERACTION}}	TRUE	Display VASP attribution details and subpoena recommendations in Section 5 & 7.
{{OVERALL_RISK_LEVEL}}	HIGH	Display urgent priority recommendations (extend tracing, asset freeze, legal notice).